Algebraic Equations and Hypergeometric Series

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Summary. We study the solutions of a general n th order algebraic equation represented by multidimensional hypergeometric series. We provide a detailed description of the domains of convergence of these series in terms of the amoeba and the Horn–Kapranov uniformization of the corresponding discriminant. From a geometric viewpoint this amounts to describing the maximal Reinhardt domains in the complement of the discriminant locus.

1 Introduction

The problem of solving algebraic equations is ubiquitous in Mathematics, and it has a long history. A formidable contribution to the development of the theory of algebraic equations was made by Niels Henrik Abel, proving in 1824 his immortal theorem on the impossibility of solving the general quintic equation by means of radicals. (The complete text [1] was published in 1826.)

This meant that Algebra had to relinquish her monopoly in the field of algebraic equations, and mathematicians turned to the idea, already proposed by Viète over two hundred years earlier, of searching for analytic solutions. The first success in this direction was achieved in 1858 by Hermite and Kronecker, see [8], who were able to express the solution to the quintic equation by means of a modular elliptic function. (This function depends on one variable, connected with the parameter a of the equation $1 + a\,x + x^5 = 0$; it was proved by Bring that any fifth degree equation can be brought into this form by means of Tschirnhaus transformations.)

Around the beginning of the last century another class of special functions made its way into the study of algebraic equations: these were the multivariate *hypergeometric* functions, given either as series [3], as integrals [10], or as solutions to differential equations [9]; see [2] for an extensive account.

Hypergeometric series are among the simplest transcendent functions, for it is possible to apply algebraic procedures to deal with them. This is inherent in their very definition as formulated by Horn in [6]: a power series in m variables is called hypergeometric if all its m ratios of subsequent coefficients are *rational* functions in the summation indices of the series. In the 1920's Birkeland published a series of papers, culminating with [3], where he presented a collection of hypergeometric series in the coefficients of the algebraic equation, which constitute solutions to the equation in certain open subsets of the coefficient space. More precisely, if one considers the general n th degree algebraic equation

$$a_0 + a_1\,x + a_2\,x^2 + \ldots + a_{n-1}\,x^{n-1} + a_n\,x^n = 0 \tag{1}$$

with complex coefficients $a = (a_0, \ldots , a_n)$, one quickly finds, see (2) below, that its root $x(a)$, viewed as a multivalued algebraic function, has a double homogeneity property; hence it may be treated as a function of only $n-1$ variables. It is therefore enough to be able to solve equations of the type

$$a_0 + a_1\,x + \ldots + x^p + \ldots + x^q + \ldots + a_{n-1}\,x^{n-1} + a_n\,x^n = 0\,, \tag{$1'$}$$

where the coefficients of any two fixed monomials have been frozen. The formulas (6) of Birkeland constitute Taylor series solutions to $(1')$ in the remaining variables $a_0, a_1, .. [p] .. [q] .. , a_n$. By means of a simple monomial substitution corresponding to the bi-homogeneity (2), these Taylor series may at will be turned into Laurent–Puiseux series solutions of the original equation (1).

The aim of our present investigation is to describe the convergence regions of Birkeland's Taylor series solutions to $(1')$, and consequently, of the Laurent–Puiseux series solutions to (1). The most basic result regarding the convergence of power series of one or several variables is Abel's lemma, which states that if a point a belongs to the domain of convergence D of a power series σ, then σ actually converges in the whole polydisk determined by a. That is, the Cartesian product of the disks $|z_\nu| < |a_\nu|$ is contained in D, and hence the domain of convergence is completely determined by the absolute values of each of the coordinates.

The qualitative structure of the domains of convergence of power series with singularities along algebraic hypersurfaces are closely related to the *amoeba* of this surface, that is, its projection on the space of absolute values in a logarithmic scale. For a power series that represents a rational or meromorphic function the domain of convergence D is given by $\mathrm{Log}^{-1}(E)$, where E is one of the connected components of the complement of the amoeba of the polar hypersurface. Here the principle that "power series converge up to the first singularity" applies in its purest form: "and no further". In our situation, where the power series will be representing the multivalued algebraic function $x\,(a)$ with branching singularities along the discriminant locus ∇ of equation (1), the picture is more complicated: here the series represents one or several of the branches of the full algebraic function $x\,(a)$, and may therefore fail to "recognize" certain pieces of the discriminant set as being part of its singularity. A combinatorial description of the domains of convergence of the Taylor series solutions to ($1'$) is given in Theorem 3, which says that the convergence domain D_{pq} of the series (6) projects onto a domain Log (D_{pq}) containing a certain collection of the connected components of the complement of the discriminant amoeba.

In order to give a more exact description of the domains of convergence, we introduce the concept of the *contour* of the amoeba, as being the set of critical values of the mapping Log defined on the discriminant locus ∇_{pq}. The contour comprises the boundary of the amoeba, but it also contains the natural continuation of the boundary pieces that enter into the amoeba itself. In this direction we are able to exhibit a parametrization of the boundary of the convergence domains D_{pq} for certain of the series (Theorem 4), and for series depending on just two variables we obtain in Sect. 5.2 an ideal answer: a description of D_{pq} by means of explicit inequalities involving "reflections" of the discriminant. We finally remark that in the course of proving our main results, we also gained new insights regarding the discriminant locus ∇_{pq}, such as a description of its singular set (Theorem 2) and a parametrization (Theorem 1), all highly inspired by the two extraordinary papers by Horn [6] and Kapranov [7].

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2 The Solutions as Hypergeometric Series

2.1 Dehomogenization. There are two obvious homogeneity relations associated with equation (1). Firstly, the solutions remain the same if the coefficient vector a is multiplied by any non-zero scalar λ. Secondly, division of the unknown x by a non-zero number λ is compensated by multiplying each coefficient a_ν with λ^ν. In other words, any root $x\,(a)$ of equation (1), considered as a function of the coefficients a, will have the twofold homogeneity property

$$x\,(\lambda_0\,a_0, \lambda_0\lambda_1 a_1, \ldots, \lambda_0\lambda_1^n a_n) = \lambda_1^{-1}\,x\,(a)\,, \quad \lambda_0, \lambda_1 \in \mathbf{C} \setminus \{0\}\,. \tag{2}$$

One may thus regard the algebraic multivalued function $x\,(a)$ as a section of a certain holomorphic line bundle.

The presence of two homogeneities means that the roots x really depend only on $n-1$ variables, and one can actually dispose of any pair of the $1+n$ variable coefficients a_ν. Indeed, let us choose any two integers $0 \le p < q \le n$, that is, we fix an integer subsegment $[p, q] \subset [0, n]$. Then we let the two complex parameters λ_0 and λ_1 satisfy

$$\lambda_0\,\lambda_1^p = a_p^{-1}\,, \quad \text{and} \quad \lambda_0\,\lambda_1^q = a_q^{-1}\,. \tag{3}$$

It is important to observe that there are exactly $q-p$ possible choices for this pair of parameters: first we let λ_1 take any of the $q-p$ values of the radical $(a_p/a_q)^{1/(q-p)}$, and then the other parameter λ_0 is uniquely determined by either of the equations (3).

We shall use the notation $[p]$ to indicate that the index p should be omitted, and similarly for q. In view of the identity (2) this means that if $x\,(a_0, ..\,[p]\,..\,[q]\,..\,, a_n)$ is a solution of the specialized equation (1′) then $\lambda_1\, x$ will solve the original equation (1) with coefficients $\tilde{a}_\nu = \lambda_0^{-1}\lambda_1^{-\nu}\, a_\nu$. Conversely, if $x\,(a)$ is a root of (1), then $\lambda_1^{-1}x$ will be a solution of (1′) with coefficients $\tilde{a}_\nu = \lambda_0\lambda_1^\nu\, a_\nu$.

Following Sturmfels [14], we also form a matrix out of the two homogeneity relations:

$$A = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 0 & 1 & 2 & \cdots & n-1 & n \end{pmatrix} .$$

The kernel of the corresponding linear mapping is of dimension $n-1$, and for any given choice of integers p and q as above, a basis for this kernel can be composed by the integer vectors

$$(q-p)\, e_\nu + (\nu - q)\, e_p + (p-\nu)\, e_q\,, \quad \nu \neq p, q\,, \tag{4}$$

with $e_0, ..., e_n$ being the standard basis vectors in $\mathbf{R}^{1+n}$. We shall denote by B_{pq} the $(n+1)\times(n-1)$-matrix

$$B_{pq} = \left(\beta_\mu^\nu\right)\,, \tag{5}$$

whose column vectors β^ν are the basis vectors (4). One clearly has $AB_{pq} = 0$, and the square matrix obtained by deleting the rows β_p and β_q numbered p and q from B_{pq}, is equal to $q-p$ times the identity matrix of size $n-1$.

2.2 Series Solutions. If we put all coefficients a_ν in equation (1′) equal to zero, then we are left with the binomial equation $x^p + x^q = 0$. Disregarding the zero solution, we find that x is equal to any one of the radicals $(-1)^{1/(q-p)}$. It follows, from the implicit function theorem for analytic functions, that for each choice ϵ of this radical there exists a power series solution to (1′) which converges for small values of the coefficients a, and equals ϵ at the origin.

There is a classical method of Lagrange that in principle allows one to find the power series expansion of an implicitly given function, and by making clever use of this method, Birkeland was able, see [3], to explicitly write down the $q-p$ solutions to equation (1′) as the following power series:

$$\sum_{k\in\mathbf{N}^{n-1}} \frac{\epsilon^{-\langle\beta_q,k\rangle+1}}{(q-p)\,k!}\,\frac{\Gamma\big((-\langle\beta_q,k\rangle+1)/(q-p)\big)}{\Gamma\big(1+(\langle\beta_p,k\rangle+1)/(q-p)\big)}\, a_0^{k_0}a_1^{k_1}\cdot\cdot[p]\cdot\cdot[q]\cdot\cdot a_n^{k_n}\,. \tag{6}$$

Here $k! = k_0!\,k_1!\cdot\cdot[p]\cdot\cdot[q]\cdot\cdot k_n!$ and β_p, β_q are row vectors of the matrix B_{pq} from (5). A word of caution is in order here: the arguments of the Γ functions occurring in (6) will sometimes be negative integers, corresponding to poles of Γ.

However, the quotient of the two Γ functions should be thought of as a meromorphic function with *removable* singularities at the integer points. For instance, we mean that $\Gamma(-1)/\Gamma(-2) = -2\,\Gamma(-2)/\Gamma(-2) = -2$.

We shall *denote this series by* σ_{pq} even though it really does depend also on a choice of radical ϵ, which coincides with its value at the origin. The particular choice of ϵ will not influence the convergence properties of σ_{pq}. Whenever $q-p=1$ the series is truly hypergeometric in the sense of Horn, meaning that the quotient of subsequent coefficients are rational functions of k. As was pointed out by Birkeland, one can regroup (6) as a sum of $(q-p)^{n-1}$ hypergeometric series by simply taking the equivalence classes of each index k_ν modulo $q-p$. We shall here for convenience be referring to σ_{pq} itself as a hypergemetric series.

Our object in this note will be to determine, as exactly as possible, the domains of convergence of the series solutions (6). What is clear from the start is that each σ_{pq} will converge in some sufficiently small polydisk around the origin. In [3] Birkeland also gave an explicit, though rather crude, sufficient condition for the convergence, formulated in terms of a bound on the sum $|a_0| + |a_1| + ..\,[p]\,..\,[q]\,.. + |a_n|$, but to the best of our knowledge no exact convergence criteria for σ_{pq} have hitherto been presented.

Let us point out that each series (6) will also give rise to a Laurent–Puiseux series solution of the original equation (1). Indeed, we just have to use the recipe given above, and write

$$\lambda_1\, \sigma_{pq}(\lambda_0\, a_0, \lambda_0\lambda_1\, a_1, ..\,[p]\,..\,[q]\,.., \lambda_0\lambda_1^n\, a_n)\ ,$$

with λ_0 and λ_1 given by (3). The fact that (6) converges for small enough $|a_\nu|$ will then be equivalent to the corresponding Laurent–Puiseux series converging for

$$|a_\nu|^{q-p}/\big(|a_p^{q-\nu}||a_q^{p-\nu}|\big) < M_\nu\,, \quad \nu = 0, 1, ..\,[p]\,..\,[q]\,.., n\ ,$$

with the positive constants M_ν being small enough.

3 The Discriminant

3.1 Definition and Basic Properties. When $n > 1$ there is intimately connected with equation (1) a remarkable polynomial Δ in the coefficients a. It is called the discriminant of the equation and it describes the occurrence of multiple roots in the sense that $\Delta\,(a) = 0$ if and only if (1) has at least one multiple root. The discriminant Δ is an irreducible polynomial with integer coefficients and may be expressed in terms of the roots $x_1, x_2, \ldots, x_n$ of the equation by the simple symmetric formula

$$\Delta = \pm\, a_n^{2(n-1)} \prod_{j<k} (x_j - x_k)^2\,.$$

We shall follow the convention of [5], Chap. 12, and choose the sign to be $(-1)^{n(n-1)/2}$.

A more explicit formula for Δ, directly in terms of the coefficients of the equation, is provided by the classical Sylvester determinant

$$\Delta = \frac{1}{a_n} \begin{vmatrix} a_0 & a_1 & a_2 & \dots & a_{n-1} & a_n & 0 & \dots \\ 0 & a_0 & a_1 & \dots & a_{n-2} & a_{n-1} & a_n & \dots \\ \vdots & \dots & & \dots & & \dots & & \vdots \\ a_1 & 2a_2 & \dots & (n-1)a_{n-1} & n\,a_n & 0 & 0 & \dots \\ 0 & a_1 & \dots & (n-2)a_{n-2} & (n-1)a_{n-1} & n\,a_n & 0 & \dots \\ \vdots & \dots & & \dots & & \dots & & \vdots \end{vmatrix} . \tag{7}$$

Here the first $n-1$ rows contain translates of the coefficient vector a, and the remaining n rows involve translates of the coefficients of the differentiated left hand side of (1). For $n = 2$ this gives $\Delta = 4\,a_0a_2 - a_1^2$, and for $n = 3$ one obtains

$$\Delta = 27\,a_0^2a_3^2 + 4\,a_1^3a_3 + 4\,a_0a_2^3 - 18\,a_0a_1a_2a_3 - a_1^2a_2^2\,.$$

The number of terms in Δ grows rapidly with increasing n, and so does the size of its coefficients. For $n = 4$ there are sixteen terms, and for $n = 5$ fifty-nine terms. For any n the monomials $n^n a_0^{n-1}a_n^{n-1}$ and $\pm a_1^2a_2^2\cdots a_{n-1}^2$ occur in Δ, and they are among the 2^{n-1} extremal terms of the discriminant for which a simple combinatorial formula will be given below.

We are going to denote by ∇ the zero locus of the discriminant Δ. Geometrically, this hypersurface ∇ is the dual variety of the Veronese curve $[1 : x : x^2 : \ldots : x^n]$ in $\mathbf{CP}^n$, that is, it consists of those vectors $a \in \mathbf{C}^{1+n}$ for which the corresponding hyperplane $a_0z_0 + a_1z_1 + \ldots + a_nz_n = 0$ is not transversal to the parametrized curve $z_\nu = x^\nu$.

The bi-homogeneity (2) of the roots has the following analog for the discriminant:

$$\Delta\,(\lambda_0\,a_0, \lambda_0\lambda_1 a_1, ..., \lambda_0\lambda_1^n a_n) = \lambda_0^{2(n-1)}\lambda_1^{n(n-1)}\Delta\,(a)\,. \tag{8}$$

Now let Δ_{pq} denote the polynomial obtained from the homogeneous discriminant Δ by simply setting a_p and a_q equal to 1. Then Δ_{pq} depends only on the remaining $n-1$ variables; in fact, Δ_{pq} is the discriminant of the equation (1′). Using again the parameters λ_0 and λ_1 that satisfy (3), we deduce from the identity (8) that

$$\Delta_{pq}(\tilde{a}_0, ..\,[p]\,..\,[q]\,.., \tilde{a}_n) = a_p^{s_p}a_q^{s_q}\;\Delta(a)\,, \tag{9}$$

where $\tilde{a}_\nu = a_\nu\,a_p^{(\nu-q)/(q-p)}a_q^{(p-\nu)/(q-p)}$ and the exponents s_p and s_q are given by $(n-2q)(n-1)/(q-p)$ and $(2p-n)(n-1)/(q-p)$ respectively.

The equation $\Delta_{pq} = 0$ can be interpreted as the homogeneous equation $\Delta = 0$ expressed in suitable local coordinates where $a_pa_q \neq 0$, and in accordance with the theory of toric manifolds the connection between these equations is given by an identity

$$\Delta_{pq}\big(a^{B_{pq}^{\mathrm{tr}}/(q-p)}\big) = a_p^{s_p}\;a_q^{s_q}\;\Delta(a)\,, \tag{10}$$

where $B^{\mathrm{tr}}_{pq}/(q-p)$ denotes the transpose of the matrix B_{pq} divided by the scalar $q-p$. Since $a_\nu a_p^{(\nu-q)/(q-p)} a_q^{(p-\nu)/(q-p)} = a^{\beta^\nu/(q-p)}$ the equality (10) is in fact the same as equation (9) above. We emphasize once more that in all these identities there are really only radicals of the single quantity a_p/a_q that produce any ambiguity.

3.2 Subdivisions, Polytopes, and Amoebas. A *subdivision* of an integer line segment $[0, n]$ is the collection of adjacent subsegments obtained by dividing the original segment at the points of some subset $I \subset \{1, 2, \dots, n-1\}$.

The *Newton polytope* of a polynomial is the convex hull of the set of exponent vectors of its monomials, with these vectors being considered as integer lattice points in the corresponding real vector space.

The *amoeba* of a polynomial is the image of its zero locus under the mapping Log, which sends each variable to the logarithm of its absolute value.

We shall let the Newton polytope and the amoeba of the discriminant Δ of equation (1) be denoted by $\mathcal{N}$ and $\mathcal{A}$ respectively. We remark in passing that the Newton polytope of the polynomial on the left hand side of (1) is equal to the line segment $[0, n]$.

It will be of importance for our investigations that *there exist natural bijections between the following sets*:

$$\big\{\text{subdivisions of } [0,n]\big\} \leftrightarrow \big\{\text{vertices of } \mathcal{N}\big\} \leftrightarrow \big\{\text{components of } \mathbf{R}^{1+n} \setminus \mathcal{A}\big\}$$

Let us explain these bijections in some detail. The first one, between subdivisions and vertices, is a special instance of a remarkable result from [5] relating triangulations of certain polytopes (in our case the segment $[0, n]$) to extremal monomials in general discriminants (in our case the classical discriminant Δ). To any given subdivision $[0, i_1], [i_1, i_2], \dots, [i_s, n]$, obtained from an integer subset $I = \{i_1 < i_2 < \cdots < i_s\}$, one assigns the vector

$$(d_0-1)e_0 + (d_0+d_1)e_{i_1} + (d_1+d_2)e_{i_2} + \ldots + (d_{s-1}+d_s)e_{i_s} + (d_s-1)e_n\,, \tag{11}$$

where $e_0, e_1, \dots, e_n$ are the standard basis vectors in $\mathbf{R}^{1+n}$, and d_ν is the length of the subsegment $[i_\nu, i_{\nu+1}]$; we set $i_0 = 0$ and $i_{s+1} = n$. This vector is one of the 2^{n-1} vertices of the Newton polytope $\mathcal{N}$, and, since each different subset I gives rise to a different vector in this way, the described assignment does indeed establish a bijection between subdivisions and vertices. In fact, the vertex monomial corresponding to I appears in Δ with the coefficient

$$\prod_{\nu=0}^{s} \big[(-1)^{i_\nu} d_\nu\big]^{d_\nu}\,.$$

For instance, when $n = 2$ the two subdivisions $[0, 2]$ and $[0, 1], [1, 2]$ correspond to the vertex vectors $(1, 0, 1)$ and $(0, 2, 0)$ respectively, and the two vertex monomials are $4\,a_0a_2$ and $-a_1^2$. In the case $n = 3$ there are four subdivisions $[0, 3]$;

[0, 1], [1, 3]; [0, 2], [2, 3]; [0, 1], [1, 2], [2, 3] with associated vertices (2, 0, 0, 2), (0, 3, 0, 1), (1, 0, 3, 0), (0, 2, 2, 0), and corresponding monomials $27\, a_0^2 a_3^2$, $4\, a_1^3 a_3$, $4\, a_0 a_2^3$, $-a_1^2 a_2^2$.

One may also describe the Newton polytope $\mathcal{N}$ by means of explicit equations and inequalities. Indeed, it is equal to the skew cube of dimension $n-1$ determined by the two homogeneity relations $t_0+t_1+\ldots+t_n = 2(n-1)$ and $t_1+2\,t_2+\ldots+n\,t_n = n(n-1)$, together with the $2(n-1)$ linear inequalities

$$t_k \geq 0\,, \quad \sum_{j=1}^{n-1} \min(j,k)\big[n-\max(j,k)\big]\, t_j \leq nk(n-k)\,, \quad k=1,2,\ldots,n-1\,.$$

The other bijection, the one between vertices of the Newton polytope and connected components of the amoeba complement, has to do with normal cones and Laurent series expansions of rational functions.

The *normal cone* C_v at the vertex v of a polytope is the closed convex polyhedral cone, having its apex at the origin, spanned by the outer normals of those faces of the polytope that come together at v. The union of the normal cones at all the vertices of the polytope equals the entire ambient vector space, and the intersection $C_v \cap C_w$ of any two normal cones is a polyhedral cone of codimension equal to the dimension of the smallest face containing the two vertices v and w. The *recession cone* of a convex set E is the maximal cone C with apex at the origin such that the vector sum $E+C$ is contained in E. If E is bounded then $C=\{0\}$.

A multidimensional *Laurent series* is a power series of the form

$$\sum_{k\in\mathbf{Z}^{1+n}} c_k\, a_0^{k_0} a_1^{k_1}\cdots a_n^{k_n}\,, \qquad c_k \in \mathbf{C}\,.$$

Its domain of convergence is given by $\mathrm{Log}^{-1}(U)$ for some convex domain $U \subset \mathbf{R}^{1+n}$. Any holomorphic function in a domain of the type $\mathrm{Log}^{-1}(U)$ can be represented by a unique Laurent series that converges in that domain. For every polynomial f there is a one-to-one correspondence between the Laurent expansions of $1/f$ and the connected components of the amoeba complement $\mathbf{R}^{1+n} \setminus \mathcal{A}_f$.

Let $\{E\}$ denote the family of connected components of the complement of $\mathcal{A}_f$. It was shown in [4] that there is an injective mapping $\nu\colon \{E\} \to \mathcal{N}_f \cap \mathbf{Z}^{1+n}$ with the property that the normal cone to $\mathcal{N}_f$ at $\nu(E)$ is equal to the recession cone of E. In particular, if the recession cone of E has a non-empty interior, then $\nu(E)$ is a vertex of the Newton polytope $\mathcal{N}_f$.

Conversely, given a vertex v of the Newton polytope $\mathcal{N}_f$, one can obtain a convergent Laurent series by identifying the vertex monomial $f_v\, a^v$ in f and forming a geometric progression

$$\frac{1}{f(a)} = \sum_{k=0}^{\infty} \frac{(f_v\, a^v - f(a))^k}{(f_v\, a^v)^{1+k}}\,.$$

The domain of convergence for the Laurent series that arises in this way will have the normal cone C_v as its recession cone. In fact, it coincides with the unique component E of $\mathbf{R}^{1+n} \setminus \mathcal{A}_f$ for which $\nu(E) = v$.

For a general polynomial f the mapping ν provides a bijection between the set of vertices of $\mathcal{N}_f$ and a *subset* of the family of complement components $\{E\}$. However, if we now specialize, and take for f a discriminant polynomial Δ, then the number of complement components is minimal and they all correspond to vertices of the Newton polytope. A proof of this fact is given in [12].

Figure 1 illustrates the amoebas of the discriminants Δ_{03} and Δ_{01} for the case $n = 3$. Notice that the boundary curves cut into the amoebas. These entire curves constitute what we call the contours of the amoebas, and will be carefully studied in connection with the Horn–Kapranov uniformization below.

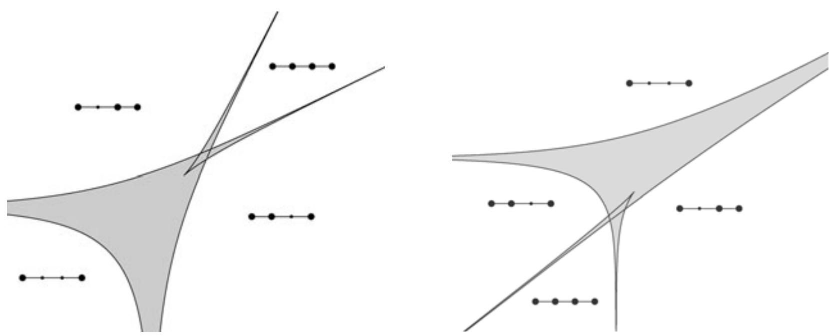

Fig. 1. The cubic [0, 3] and [0, 1] discriminant amoebas (*shaded*), their contours (*solid*), and the subdivisions associated with the complement components

3.3 Uniformization of the Discriminant Locus. By means of the matrix B_{pq} we now define a multivalued algebraic mapping

$$\psi_{pq} \colon \mathbf{CP}^{n-2} \to \mathbf{C}^{n-1}$$

from projective space into the space of coefficients $(a_0, ..[p]..[q].., a_n)$ of the equation $(1')$, by setting

$$\psi_{pq}(s_1 : s_2 : \ \ldots : s_{n-1}) = \left(\prod_{j=0}^{n} \langle \beta_j, s \rangle^{\beta_j^0/(q-p)}, ..[p]..[q].., \prod_{j=0}^{n} \langle \beta_j, s \rangle^{\beta_j^n/(q-p)} \right),$$

with the vectors $\beta_0, \ldots, \beta_n$ being the row vectors of the matrix B_{pq}. More precisely, in view of (4) we may write the ν th component of ψ_{pq} as

$$\left[\frac{\langle \beta_p, s \rangle}{\langle \beta_q, s \rangle} \right]^{\frac{\nu-p}{q-p}} \frac{\langle \beta_\nu, s \rangle}{\langle \beta_p, s \rangle},$$

and thus the multi-valuedness arises only from the choice of a radical $\left[\langle\beta_p, s\rangle/\langle\beta_q, s\rangle\right]^{1/(q-p)}$. It follows that ψ_{pq} is a one-to-$(q-p)$ mapping.

The following theorem generalizes (a special instance of) a result of Kapranov, who proved the case $q-p=1$ in [7]. He called ψ_{pq} the Horn uniformization, since it first occurred in the paper [6] in connection with the convergence of hypergeometric series. Recall that the *logarithmic Gauss mapping* of a hypersurface $\nabla = \{f(a)=0\}$ is defined as the mapping γ from ∇ into projective space with components given by $\gamma_\nu(a) = a_\nu\,\partial f(a)/\partial a_\nu$. Geometrically γ is obtained (locally) by taking the image of ∇ under the coordinate-wise mapping $a_\nu \mapsto \log a_\nu$, and then letting $\gamma(a)$ be the normal line to $\log(\nabla)$ at the point $\log a$. We use the same notation γ for the Gauss mapping of any of the discriminant hypersurfaces ∇_{pq}.

Theorem 1. *The mapping ψ_{pq} parametrizes the discriminant hypersurface*

$$\nabla_{pq} = \left\{\Delta_{pq} = 0\right\}.$$

Moreover, it is inverse to the logarithmic Gauss mapping $\gamma\colon \nabla_{pq} \to \mathbf{CP}^{n-2}$, which hence realizes the discriminant hypersurface ∇_{pq} as a $(q-p)$-fold covering of the projective space $\mathbf{CP}^{n-2}$.

Proof. Let us first show that $\psi_{pq}(\mathbf{CP}^{n-2}) \subset \nabla_{pq}$. To this end, we begin by observing that

$$\Delta(\langle B_{pq}, s\rangle) = 0\,,$$

with $\langle B_{pq}, s\rangle$ denoting the vector whose components are

$$\langle\beta_0, s\rangle, \langle\beta_1, s\rangle, \ldots, \langle\beta_n, s\rangle\,.$$

Indeed, in the Sylvester matrix (7) for the polynomial $\Delta(\langle B_{pq}, s\rangle)$ there are $n-1$ rows containing translates of the vector $\langle B_{pq}, s\rangle$ with the remaining entries being equal to zero, and the other n rows similarly all consist of translates of

$$\langle\beta_1, s\rangle, 2\langle\beta_2, s\rangle \ldots, n\langle\beta_n, s\rangle\,.$$

But since $AB_{pq} = 0$, this means that the Sylvester matrix annihilates the vector $(1, 1, \ldots, 1)$, and hence the Sylvester determinant equals zero, which precisely says that $\Delta(\langle B_{pq}, s\rangle) = 0$ as claimed. In view of formula (10) we therefore get the desired vanishing

$$\Delta_{pq}\left(\langle B_{pq}, s\rangle^{B^{\mathrm{tr}}/(q-p)}\right) = 0\,.$$

Our next step will be to prove that the mapping $\tilde{\psi}_{pq}$, obtained from ψ_{pq} by raising each component to the power $q-p$, is a birational isomorphism from $\mathbf{CP}^{n-2}$ onto some hypersurface $\widetilde{\nabla}_{pq} \subset \mathbf{C}^{n-1}$, and furthermore, that its inverse is equal to the logarithmic Gauss mapping $\tilde{\gamma}$ of $\widetilde{\nabla}_{pq}$. The fact that the image $\widetilde{\nabla}_{pq}$ of the mapping $\tilde{\psi}_{pq}$ is indeed a hypersurface follows by checking that the Jacobian of $\tilde{\psi}_{pq}$ is generically non-zero. This is easily done, for instance by making a computation in the local coordinates $(1, s_1, s_2, .. [p] .. [q] .. , s_n)$ on $\mathbf{CP}^{n-2} \setminus \{s_0 = 0\}$, in terms of which $\tilde{\psi}_{pq}$ has the form

$$\tilde{\psi}_{pq} = \left(\varphi_0, s_1^{q-p}\varphi_1, .. [p] .. [q] .. s_n^{q-p}\varphi_n\right),$$

with $\varphi_j(0) \neq 0$ because the row vectors β_p and β_q contain no zero coordinates. To see that $\tilde{\psi}_{pq}$ is inverse to the logarithmic Gauss mapping we use the symmetry relations

$$\frac{\partial \log \psi_k}{\partial s_j} = \frac{\partial \log \psi_j}{\partial s_k}, \quad j, k = 0, 1, .. [p] .. [q] .. , n ,$$

which hold for both for ψ_{pq} and $\tilde{\psi}_{pq}$. They imply that the Jacobian matrix J of the mapping $\mathrm{Log} \circ \tilde{\psi}_{pq}$ is symmetric, and that it is annihilated by the vector s. Hence the composition $\tilde{\gamma} \circ \tilde{\psi}_{pq}$ is indeed the identity mapping. Similary, we see that also $\gamma \circ \psi_{pq}$ reduces to the identity. This argument is the same as in [7] (the proof of the second half of part b) of Theorem 2.1).

All that now remains to be shown is that ψ_{pq} maps $\mathbf{CP}^{n-2}$ onto the discriminant hypersurface ∇_{pq}. Indeed, in analogy with the mapping $\tilde{\psi}_{pq}$ we see that each of the $q-p$ branches of ψ_{pq} is a local immersion. Hence the image of ψ_{pq} is a hypersurface. On the other hand, we know that the discriminant hypersurface $\nabla = \{\Delta = 0\}$ is irreducible, and in view of the identity (10) the hypersurface ∇_{pq} is also irreducible. From this we conclude that the image of ψ_{pq} is equal to ∇_{pq} as required. Theorem 1 is proved. ■

Let us take a closer look at the relation between the hypersurfaces $\widetilde{\nabla}$ and ∇. To this end we consider the power mapping $F\colon \mathbf{C}^{n-1} \to \mathbf{C}^{n-1}$ given by

$$F_\nu(a) = a_\nu^{q-p}, \quad \nu \neq p, q ,$$

and examine the full preimage $F^{-1}(\widetilde{\nabla}_{pq})$. Since $\tilde{\psi}_{pq} = F \circ \psi_{pq}$ we see that $\nabla_{pq} \subset F^{-1}(\widetilde{\nabla}_{pq})$. When $q - p = 1$ there is equality $\nabla_{pq} = F^{-1}(\widetilde{\nabla}_{pq}) = \widetilde{\nabla}_{pq}$, but in general the preimage of $\widetilde{\nabla}_{pq}$ under F will contain a number of "doubles" of the true discriminantal locus ∇_{pq}. This is made precise in the following proposition.

Proposition 1. *The preimage $F^{-1}(\widetilde{\nabla}_{pq})$ consists of $(q-p)^{n-2}$ irreducible components each having the same amoeba $\mathcal{A}_{pq}$ as the true discriminant hypersurface ∇_{pq}.*

Proof. We obtain the full preimage by composing ψ_{pq} with F, and then taking the $(q-p)$ th root of each component separately:

$$\left(\psi_{pq}^{q-p}\right)^{1/(q-p)} = \left\{\left(\omega^{j_0}\alpha\, a_0/a_p, .. [p] .. [q] .. , \omega^{j_n}\alpha\, a_n/a_p\right)\right\} .$$

Here $\omega = \exp 2\pi i/(q-p)$, and α is a fixed value of the radical $(a_p/a_q)^{1/(q-p)}$, while the indices j_k run independently over the set $[0, q-p-1]$. The entire family of such multi-indices $(j_0, ..[p]..[q].., j_n)$ splits into $(q-p)^{n-2}$ groups, each consisting of $q-p$ multi-indices, which constitute cycles for the monodromy of the function $\tilde{\psi}^{1/(q-p)}$. Representatives for these cycles can for instance be obtained by fixing some $\nu \neq p, q$, and considering the set J_ν consisting of those multi-indices that satisfy $j_\nu = 0$. This means that we can write

$$F^{-1}(\widetilde{\nabla}_{pq}) = \left\{ \prod_{J_\nu} \Delta_{pq}(\omega^{j_0} a_0, ..[p]..[q].., \omega^{j_n} a_n) = 0 \right\} . \tag{12}$$

The factor corresponding to the zero multi-index is the true discriminant Δ_{pq}, and it is clear that all the other factors will also have the same amoeba. ■

We are now going to consider the restriction of the uniformization map ψ_{pq} to the real space $\mathbf{RP}^{n-2}$, and we introduce the following new notion.

Definition. *The set of critical values of the mapping* $\operatorname{Log}\colon \nabla_{pq} \to \mathbf{R}^{n-1}$ *is called the* contour *of the amoeba* $\mathcal{A}_{pq}$, *and it is denoted* $\mathcal{C}_{pq}$.

Observe in particular that the boundary $\partial\mathcal{A}_{pq}$ of the amoeba is contained in the contour $\mathcal{C}_{pq}$.

Proposition 2. *The contour* $\mathcal{C}_{pq}$ *of the amoeba* $\mathcal{A}_{pq}$ *coincides with the image of* $\mathbf{RP}^{n-2}$ *under the composed mapping* $\operatorname{Log}\circ\psi_{pq}$

Proof. We consider first the single-valued mapping $\tilde{\psi}_{pq}$. It is a known fact, see for instance [11] or [15], that the set of critical points for the mapping Log is equal to $\gamma^{-1}(\mathbf{RP}^{n-2})$, and the same relation holds true for corresponding mappings $\widetilde{\operatorname{Log}}$ and $\tilde{\gamma}$ defined on $\widetilde{\nabla}_{pq}$. In fact, since the image of $\mathbf{RP}^{n-2}$ under the mapping $\tilde{\psi}_{pq}$ is contained in the real part of $\widetilde{\nabla}_{pq}$, we actually have that

$$\tilde{\psi}_{pq}\left(\mathbf{RP}^{n-2}\right) \;=\; \widetilde{\nabla}_{pq} \cap \mathbf{R}^{n-1} \;=\; \operatorname{Crit}\left(\widetilde{\operatorname{Log}}\right) .$$

Since the mappings $\operatorname{Log}\circ F^{-1}$ and $\widetilde{\operatorname{Log}}$ are identical, the proof is complete. ■

$$\begin{tikzcd}
 & \widetilde{\nabla}_{pq} \arrow[dl, bend left, "\tilde{\gamma}"] \arrow[dr, "\widetilde{\operatorname{Log}}"] & \\
\mathbf{CP}^{n-2} \arrow[ur, bend left=40, "\tilde{\psi}_{pq}"] \arrow[r, bend left, "\psi_{pq}"] & \nabla_{pq} \arrow[u, "F"] \arrow[l, bend left, "\gamma"] \arrow[r, "\operatorname{Log}"] & \mathcal{A}_{pq} \subset \mathbf{R}^{n-1} \\
\cup & \cup & \cup \\
\mathbf{RP}^{n-2} \arrow[r, "\psi_{pq}"] & X_{pq} \arrow[r, "\operatorname{Log}"] & \mathcal{C}_{pq} \subset \mathbf{R}^{n-1}
\end{tikzcd}$$

Fig. 2. The Horn–Kapranov parametrization ψ_{pq}, its interaction with the Gauss mapping γ, the Log mapping, and the contour $\mathcal{C}_{pq}$ of the amoeba $\mathcal{A}_{pq}$

3.4 Singularities of the Discriminant Locus. In this section we will show that the singular part of the discriminant hypersurface ∇_{pq} is parametrized by the restriction of the Horn–Kapranov mapping ψ_{pq} to a certain projective hyperplane. Recall that the row vectors β_p and β_q of the matrix B_{pq} have no coordinates equal to zero. They can thus be viewed as elements of the complex torus group $\mathbf{T}^{n-1} = \big(\mathbf{C} \setminus \{0\}\big)^{n-1}$, and we let $\beta_p\beta_q$ denote their product in this group:

$$\beta_p\beta_q = (\beta_{p0}\beta_{q0}, ..[p]..[q].., \beta_{pn}\beta_{qn}) \; .$$

Theorem 2. *The singular part of the discriminant hypersurface ∇_{pq} consists of the image $\psi_{pq}(L_{pq})$ of the projective hyperplane*

$$L_{pq} \;=\; \big\{\langle \beta_p\beta_q, s\rangle = 0\big\} \;\subset\; \mathbf{CP}^{n-2} \; .$$

Consequently, the singular part of the amoeba contour $\mathcal{C}_{pq}$ consists of the image $\psi_{pq}(L_{pq} \cap \mathbf{RP}^{n-2})$ of the corresponding real projective hyperplane.

Proof. It will be convenient to relabel the components of the mapping ψ_{pq} as $\psi_1, \psi_2, \ldots, \psi_{n-1}$, and to consider it as a homogeneous mapping defined on $\mathbf{C}^{n-1} \setminus \{0\}$. We must show that the rank of the Jacobian matrix $\big(\partial\psi_k/\partial s_j\big)$ drops below $n-2$ precisely on the hyperplane L_{pq}, and we begin by observing that it is equivalent to prove that the corresponding rank property holds for the logarithmic Jacobian matrix

$$J \;=\; \big(\partial \log \psi_k/\partial s_j\big)\,, \qquad j, k = 1, 2, \ldots, n-1\,.$$

Note that the homogeneity of $\log \psi_k$ implies that we have an Euler formula $\sum_{j=1}^{n-1} s_j\, \partial\psi_k/\partial s_j = 0$, and that the determinant of the full matrix J therefore vanishes.

What has to be shown is thus that the common zero locus of all the $(n-2)$-minors of J is equal to the hyperplane L_{pq}. This will now be achieved by way of the following proposition, which provides explicit formulas for all these minors. ■

Proposition 3. *The $(n-2)$-minor J_{jk} of the matrix J, obtained by excluding the j th row and the k th column, is equal to*

$$J_{jk} \;=\; \frac{(-1)^{j+k}\, s_j s_k \langle \beta_p\beta_q, s\rangle}{s_1 s_2 \cdots s_{n-1} \langle \beta_p, s\rangle \langle \beta_q, s\rangle}\;. \tag{13}$$

Proof. In order to avoid double indexing we will temporarily rewrite the two rows β_p and β_q of the matrix B_{pq} as $(p_1, \ldots, p_{n-1})$ and $(q_1, \ldots, q_{n-1})$ respectively. We also write $P(s)$ and $Q(s)$ for the corresponding linear forms $\langle \beta_p, s\rangle$ and $\langle \beta_q, s\rangle$.

The minor $J_{n-1,1}$ equals the coefficient of $ds_1 \wedge \cdots \wedge ds_{n-2}$ in the differential form $d\log\psi_2 \wedge \cdots \wedge d\log\psi_{n-1}$, and keeping in mind the identities $\log\psi_k = \log s_k + p_k \log P(s) + q_k \log Q(s)$ and $dP \wedge dP = dQ \wedge dQ = 0$, it is straightforward to check that this coefficient is equal to

$$(-1)^{n-1}\left[\frac{p_1p_{n-1}}{s_2\cdots s_{n-2}P(s)}+\frac{q_1q_{n-1}}{s_2\cdots s_{n-2}Q(s)}+\sum_{j=2}^{n-2}\frac{\delta_{j1}\delta_{j,n-1}}{s_2\cdot\cdot[j]\cdot\cdot s_{n-2}P(s)Q(s)}\right],$$

where $\delta_{\mu\nu}$ denotes the determinant $p_\mu q_\nu - p_\nu q_\mu$. We can therefore write $J_{n-1,1}$ as $-N(s)/\big(s_2\cdots s_{n-2}P(s)Q(s)\big)$ with the linear numerator $N(s)$ given by

$$(p_1q_1q_{n-1}+p_1p_{n-1}q_1)\,s_1+\sum_{j=2}^{n-2}(\delta_{j1}\delta_{j,n-1}+p_jq_1q_{n-1}+p_1p_{n-1}q_j)\,s_j$$
$$+\,(p_{n-1}q_1q_{n-1}+p_1p_{n-1}q_{n-1})\,s_{n-1}.$$

Using the relations $p_\nu + q_\nu = -1$, which stem from the fact that the elements in each column of B sum to zero, it is now an easy task to rewrite the coefficient of s_ν in $N(s)$ as $-p_\nu q_\nu$, and hence we have verified the identity (13) for the case $(j,k)=(n-1,1)$.

A completely analogous argument will establish (13) for any (j,k) with $j\neq k$, the point being that one still immediately arrives at a linear numerator similar to $N(s)$. In order to settle the remaining diagonal cases J_{kk} we profit from the known vanishing of $\det J$ to get a linear relation

$$\sum_{j=1}^{n-1}(-1)^{j+k}\,\frac{\partial\log\psi_k}{\partial s_j}\,J_{jk}=0\,,$$

which forces (13) to hold true also for J_{kk}. This finishes the proof of Proposition 3. ■

4 Domains of Convergence

Having gathered the necessary information on the discriminant and its amoeba in the previous section, we are now ready to deal with the problem of finding the domains of convergence of the various hypergeometric series σ_{pq}. We begin with a quite accurate combinatorial description of the convergence domains in terms of the connected components of the amoeba complements $\mathbf{R}^{n-1}\setminus\mathcal{A}_{pq}$.

Theorem 3. *The domain of convergence D_{pq} of the series* (6) *is a complete Reinhardt domain with the property that the corresponding convex domain* $\operatorname{Log}(D_{pq})$ *contains all the connected components of the amoeba complement $\mathbf{R}^{n-1}\setminus\mathcal{A}_{pq}$ that are associated with subdivisions of $[0,n]$ containing the segment $[p,q]$, while it is disjoint from all the other complement components.*

Proof. We know from the implicit function theorem that D_{pq} is not empty, and Abel's lemma tells us that whenever a point a belongs to D_{pq}, then so does the full polydisk determined by a. Therefore D_{pq} is indeed a complete Rienhardt domain, and the corresponding domain $\mathrm{Log}\,(D_{pq})$ will contain the negative orthant $-\mathbf{R}_+^{n-1}$ in its recession cone. In fact, by looking at the restriction of the series (6) to the coordinate axes, and using the fact that any non-confluent hypergeometric series of one variable has a finite radius of convergence, we see that, conversely, the domain $\mathrm{Log}\,(D_{pq})$ must be contained in some translate of the negative orthant. In other words, the recession cone of $\mathrm{Log}\,(D_{pq})$ is exactly equal to $-\mathbf{R}_+^{n-1}$.

Let E be a connected component of $\mathbf{R}^{n-1} \setminus \mathcal{A}_{pq}$ that intersects the domain $\mathrm{Log}\,(D_{pq})$. Then we claim that we actually have an inclusion $E \subset \mathrm{Log}\,(D_{pq})$. Indeed, the holomorphic function $x\,(a)$ is well defined but a priori multi-valued in the domain $\mathrm{Log}^{-1}(E)$. One of its branches is given by the series (6), and hence admits a single-valued continuation to D_{pq}. Now, since the larger domain $D_{pq} \cup \mathrm{Log}^{-1}(E)$ is simply connected, the same branch remains in fact single valued on all of $\mathrm{Log}^{-1}(E)$, and it will thus have a convergent Laurent series expansion in this domain. But since this expansion coincides with the series (6) in the smaller domain $D_{pq} \cap \mathrm{Log}^{-1}(E)$, we conclude that the series (6) is actually convergent in all of $\mathrm{Log}^{-1}(E)$, and hence $E \subset \mathrm{Log}\,(D_{pq})$ as claimed.

It follows, from what we have proved so far, that the domain $\mathrm{Log}\,(D_{pq})$ cannot intersect any component of the amoeba complement $\mathbf{R}^{n-1} \setminus \mathcal{A}_{pq}$ whose recession cone is not contained in the negative orthant. On the other hand, every connected component of $\mathbf{R}^{n-1} \setminus \mathcal{A}_{pq}$ with a recession cone contained in $-\mathbf{R}_+^{n-1}$ will necessarily intersect, and hence be contained in, the domain $\mathrm{Log}\,(D_{pq})$. The following proposition therefore suffices to make the proof of Theorem 3 complete. ■

Proposition 4. *The normal cone at a vertex of the Newton polytope $\mathcal{N}_{pq}$ is contained in the negative orthant $-\mathbf{R}_+^{n-1}$ if and only if the corresponding subdivision of* $[0, n]$ *contains the segment* $[p, q]$. *In fact, the union of such normal cones is equal to* $-\mathbf{R}_+^{n-1}$.

Proof. It will be enough to show that for any vertex v, one can find a good vertex w, that is, a vertex corresponding to a subdivision that includes $[p, q]$, such that the translated cone $w + \mathbf{R}_+^{n-1}$ contains v, and further, that no translate of $\mathbf{R}_+^{n-1}$ contains more than one good vertex.

To see this, we start by letting v be a bad vertex corresponding to a subdivision $[0, i_1], [i_1, i_2], \dots , [i_s, n]$, which does not include the segment $[p, q]$. Then we can make it into a good subdivision by deleting all i_ν strictly between p and q, and adding p and q, if necessary. Recalling the assignment (11), we see that the new vertex w that we obtain in this way will have all its coordinates less than, or equal to, the ones of the vertex v that we started with. The cone $w + \mathbf{R}_+^{n-1}$ thus contains the vertex v.

On the other hand, we observe that all the good vertices span a face of the polytope $\mathcal{N}_{pq}$. To be precise, this face is given by the intersection of $\mathcal{N}_{pq}$ with the affine space

$$
\begin{aligned}
p\,t_0 + (p-1)\,t_1 + \ldots + t_{p-1} &= p\,(p-1)\,, \\
t_{p+1} = t_{p+2} = \ldots = t_{q-1} &= 0\,, \\
t_{q+1} + 2\,t_{q+2} + \ldots + (n-q)\,t_n &= (n-q)(n-q-1)\,.
\end{aligned} \tag{14}
$$

Here the first equation becomes vacuous when $p = 0$, and similarly for the last one when $q = n$. Since all coefficients in the equations (14) are positive, it is clear that no translate of $\mathbf{R}_+^{n-1}$ can contain more than one good vertex. ■

In the cases when $[p, q]$ is equal to $[0, n]$, $[0, 1]$ or $[n-1, n]$ we can determine the domains of convergence completely. This is the content of the following theorem. The result for $[0, n]$ was conjectured in [13], and proved there for $n = 3$.

Theorem 4. *For $[p, q] = [0, n]$, $[0, 1]$ or $[n-1, n]$, the restriction of* $\operatorname{Log} \circ \psi_{pq}$ *to* $\mathbf{RP}_+^{n-2}$ *parametrizes the boundary of the logarithmic convergence domain* $\operatorname{Log}(D_{pq})$ *for the series σ_{pq}.*

Proof. For simplicity we will denote the mapping ψ_{pq} and its components simply by $\psi = (\psi_1, \ldots, \psi_{n-1})$ throughout this proof. We will also consider it as a homogeneous mapping defined on $\mathbf{C}^{n-1} \setminus \{0\}$. In view of Horn's theorem on the convergence of hypergeometric series, it will be enough to prove that the surface $\operatorname{Log} \circ \psi(\mathbf{R}_+^{n-1})$ is convex, that is, that it lies "below" each of its tangent planes. Fix a point $s^0 \in \mathbf{R}_+^{n-1}$. The tangent space at the corresponding point $\operatorname{Log} \circ \psi(s^0)$ is then given by $\langle t, s^0 \rangle = 0$. What we shall prove is thus that for any s the scalar λ determined by

$$
\langle \lambda\, s + \operatorname{Log} \circ \psi(s) - \operatorname{Log} \circ \psi\left(s^0\right)\,,\; s^0\, \rangle \;=\; 0
$$

is necessarily positive. In other words, we must show that the mapping $s \mapsto -\langle \operatorname{Log} \circ \psi(s), s^0 \rangle$ has a minimum at $s = s^0$. The Hessian matrix of this mapping at the point s^0 is equal to

$$
\left[-\sum_i s_i^0\, \frac{\partial^2 \log \psi_i(s^0)}{\partial s_j \partial s_k} \right] = \left[-\sum_i s_i^0\, \frac{\partial}{\partial s_i} \left(\frac{\partial \log \psi_k(s^0)}{\partial s_j} \right) \right], \tag{15}
$$

and we claim that this is nothing but $J(s^0)$, with J denoting the Jacobian matrix $\left(\partial \log \psi_k / \partial s_j\right)$ that we encountered earlier. Indeed, this follows from (15) and the Euler identity, since the function $\partial \log \psi_k / \partial s_j$ is homogeneous of degree -1.

We know that J has one eigenvalue equal to zero, with the associated eigenvector s, and we want to show that for s in the open positive orthant all the remaining $n-2$ eigenvalues are strictly positive. Since the signature of J will vary continuously with s, it will remain unchanged as long as each $(n-2)$-minor J_{jk} stays non-zero and finite. In view of (13) this is violated only if one of the hyperplanes $\langle \beta_p, s \rangle = 0$, $\langle \beta_q, s \rangle = 0$, or $\langle \beta_p \beta_q, s \rangle = 0$ intersects the open positive orthant. Let us check that this does not happen in our cases. Indeed, for the case $[0, n]$ we get

$$
\beta_p = -(n-1, n-2, \ldots, 1)\,, \quad \beta_q = -(1, 2, \ldots, n-1)\,,
$$

while in the case [0, 1] we have

$$\beta_p = (1, 2, \dots, n-1)\ , \quad \beta_q = -(2, 3, \dots, n)\ ,$$

and similarly for $[n-1, n]$.

All that remains is to verify that for some vector s in the positive orthant, say $s = (1, 1, \dots, 1)$, all the $n-2$ non-zero eigenvalues of $J(s)$ are stricly positive. Letting I denote the diagonal unit matrix, we can compute the characteristic polynomial of $J(s)$ as $\det(J - \lambda I)$. Looking first at the case $[0, n]$, we see that $J - \lambda I$ is equal to

$$(n-\lambda)\, I - \frac{2}{n(n-1)} \beta_p^{\mathrm{tr}} \beta_p - \frac{2}{n(n-1)} \beta_q^{\mathrm{tr}} \beta_q\ .$$

Since the last two matrices in this expression have rank one, it is a straightforward matter to compute the determinant, and as a result one sees that the characteristic polynomial equals $\pm\lambda(\lambda - n)^{n-3}(\lambda - 2(n+2)/3)$. This shows that all eigenvalues are positive as desired.

Similar computations for the cases [0, 1] and $[n-1, n]$ lead to the characteristic polynomial $\pm\lambda(\lambda-1)^{n-3}(\lambda-4(n+1)/3(n+2))$, and again we see that no eigenvalues are negative. The proof is complete. ∎

5 Fewnomial Equations

5.1 Trinomials. In case our equation contains only three non-zero monomials, the procedure of reducing the number of parameters, by choosing a subsegment $[p, q]$ and putting $a_p = a_q = 1$, will leave us with a single remaining variable coefficient. The corresponding series solutions are thus one variable hypergeometric series, and their common domain of convergence is a disk. More precisely, suppose that the equation has the form

$$a_0 + a_m\, x^m + a_n\, x^n = 0\ ,$$

with the two integer exponents $0 < m < n$ being relatively prime. The discriminant of this equation is then equal to the binomial

$$\Delta = n^n a_0^{n-m} a_n^m + (-1)^{m(n-m)} m^m (n-m)^{n-m} a_m^n\ .$$

It is obtained from the discriminant of the full nth degree equation by making the specialization $a_\nu = 0$ for $\nu \notin \{0, m, n\}$, and dividing out the common factor $a_0^{m-1} a_n^{n-m-1}$. Since m and n are not both even, the sign factor $(-1)^{m(n-m)}$ may be written $(-1)^{1+n}$.

As an illustration of our general approach we now choose the subsegment $[p, q] = [0, m]$. The corresponding inhomogeneous discriminant is given by $\Delta_{0m} = n^n a_n^m - (-1)^n m^m (n-m)^{n-m}$, and its zero locus ∇_{0m} consists of m points which are "parametrized" by the trivial Horn map

$$\psi : 1 \mapsto \left[\frac{m^m (n-m)^{n-m}}{(-n)^n} \right]^{1/m} .$$

The amoeba of Δ_{0m} is equal to the single point $\log|\psi(1)|$, and the connected component $\log|a_n| < \log|\psi(1)|$ of the complement of the amoeba is the one that corresponds to the partition $[0, m, n]$. Setting $a_n = a$, we can write down the m series solutions as

$$x(a) = \sum_{k=0}^{\infty} \epsilon^{1+nk} \frac{\Gamma\big((1+nk)/m\big)}{\Gamma\big(1+(1+(n-m)k)/m\big)} \frac{a^k}{k!} \Big/ \mathrm{m}$$

with ϵ being any m th root of -1. The radius of convergence of these series is $|\psi(1)|$, that is, the exponential of the amoeba point.

In particular, if $m = 1$ and $n = 2$ we recover the solution

$$x(a) = -\sum_{k=0}^{\infty} \frac{(2k)!}{(k+1)!} \frac{a^k}{k!} = \frac{-1+\sqrt{1-4a}}{2a}$$

to the quadratic equation $1 + x + a\,x^2 = 0$, whereas the choice $m = 1$ and $n = 3$ yields the root

$$\begin{aligned} x(a) &= -\sum_{k=0}^{\infty} (-1)^k \frac{(3k)!}{(2k+1)!} \frac{a^k}{k!} \\ &= -3 \Big/ \left\{ 1 + \left[\sqrt{1+\frac{27a}{4}} + \sqrt{\frac{27a}{4}} \right]^{2/3} + \left[\sqrt{1+\frac{27a}{4}} - \sqrt{\frac{27a}{4}} \right]^{2/3} \right\} \end{aligned}$$

of the cubic equation $1 + x + a\,x^3 = 0$.

5.2 Tetranomials. The next simplest case is the one where the left hand side of the equation consists of four monomials, that is, of a tetranomial. Here the dehomogenization procedure leads to a situation with two varying coefficients a_j and a_k. This means that for each choice of a subsegment $[p, q]$, the corresponding hypergeometric series solutions, as well as the inhomogeneous discriminant and its amoeba will all depend on these two variables a_j and a_k. We are now going to see that in the tetranomial case, which in particular includes the general cubic equation, the domains of convergence of the hypergeometric series solutions admit particularly satisfactory descriptions, not only in terms of the Horn uniformization and the amoeba, but also more directly by means of explicit polynomial inequalities related to the discriminant.

The general tetranomial equation is of the form

$$a_0 + a_\ell\, x^\ell + a_m\, x^m + a_n\, x^n = 0 \,, \tag{16}$$

with three relatively prime integer exponents $0 < \ell < m < n$. Its discriminant Δ is homogeneous of degree $n + m - \ell$ and has the further homogeneity property $\Delta(a_0, \lambda^\ell a_\ell, \lambda^m a_m, \lambda^n a_n) = \lambda^{mn}$.

Let us first deal with the cubic case $\ell = 1$, $m = 2$, $n = 3$. If we choose $[p, q] = [0, 1]$ and use the inhomogeneous coordinate $(1, s)$ on $\mathbf{CP}^1$, then we explicitly have

$$B_{01} = \begin{pmatrix} 1 & 2 \\ -2 & -3 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \text{and} \quad \psi_{01}(s) = \left[\frac{1+2s}{(2+3s)^2}, -\frac{s\,(1+2s)^2}{(2+3s)^3} \right].$$

We know that $\psi_{01}(\mathbf{RP}^1) = \nabla_{01} \cap \mathbf{R}^2$, and also that $\operatorname{Log} \circ \psi_{01}(\mathbf{RP}^1) = \mathcal{C}_{01}$. By Horn's theorem it follows that the domain of convergence D_{01} for the series σ_{01} is determined by the part of the contour $\mathcal{C}_{01}$ given by $\operatorname{Log} \circ \psi_{01}(\mathbf{RP}^1_+)$. The corresponding part $\psi_{01}(\mathbf{RP}^1_+)$ of the real discriminant curve $\nabla_{01} \cap \mathbf{R}^2$ is located in the fourth quadrant. In fact, each of the four parts of $\nabla_{01} \cap \mathbf{R}^2$ that correspond to the parameter intervals $(-\infty, -2/3)$, $(-2/3, -1/2)$, $(-1/2, 0)$ and $(0, \infty)$ lie in different quadrants. This means that the boundary of D_{01} is given by the equation $\Delta_{01}(|a_2|, -|a_3|) = 0$. Looking more carefully, we can determine on which side of this surface the domain is located, and we find that

$$D_{01} = \left\{ (a_2, a_3) \in \mathbf{C}^2 ;\ 27\,|a_3|^2 - 4\,|a_3| + 4\,|a_2|^3 + 18\,|a_2||a_3| - |a_2|^2 \ < \ 0 \right\}.$$

A similar study of the other cases $[p, q]$ reveals the complete list of convergence domains:

$$\begin{aligned} D_{02} &= \left\{\Delta_{02}(|a_1|, |a_3|) > 0\right\} \cap \left\{\Delta_{02}(|a_1|, -|a_3|) < 0\right\}, \\ D_{03} &= \left\{\Delta_{03}(-|a_1|, -|a_2|) > 0\right\}, \\ D_{12} &= \left\{\Delta_{12}(|a_0|, -|a_3|) < 0\right\} \cap \left\{\Delta_{12}(-|a_0|, |a_3|) < 0\right\}, \\ D_{13} &= \left\{\Delta_{13}(|a_0|, |a_2|) > 0\right\} \cap \left\{\Delta_{13}(|a_0|, -|a_2|) < 0\right\}, \\ D_{23} &= \left\{\Delta_{23}(|a_0|, -|a_1|) < 0\right\}. \end{aligned}$$

We can use the same method to treat a general tetranomial equation (16), but in order to obtain precise discriminant inequalities as in the cubic case, we need to assume that ℓ is odd, m is even, and n is odd. Otherwise the four parts of the amoeba contour will not correspond to parts of the real discriminant curve in all the different quadrants. Let us be specific, and consider the quintic equation

$$1 + x + a\,x^2 + b\,x^5 = 0\,. \tag{17}$$

The discriminant of this equation is equal to $3125\,b^2 + 256\,b + 108\,a^5 - 27\,a^4 - 1600\,ab + 2250\,a^2 b$, and the hypergeometric series solution σ_{01} to (17) is given by

$$\sigma_{01} = -\sum_{j,k \geq 0} (-1)^k \frac{(2j+5k)!}{j!k!(j+4k+1)!}\, a^j b^k\,.$$

The domain of convergence D_{01} of this power series is now exactly described, in analogy with the corresponding series in the cubic case, by the single algebraic inequality

$$3125\,|b|^2 - 256\,|b| + 108\,|a|^5 - 27\,|a|^4 + 1600\,|a||b| - 2250\,|a|^2|b| \;<\; 0\,.$$

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